

Supplementary Material

A schedule-driven framework for estimating infectious disease importation risk at mass gathering events: the 2026 FIFA World Cup as a case study

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Appendix A. Mathematical specification of travel flow estimates

$N_{c,v}$

A.1 Notation

Symbol	Definition
c	Source country ($c = 1, \dots, C$)
v	US venue city ($v = 1, \dots, 11$)
A_c	June 2024 COR arrivals from country c to the US (I-94 programme, CBP)
$T_{c,v}$	Mean June passengers from country c to US city v , averaged over June 2023–2025 (BTS T-100 Form 41 data)
$f_{c,v}^T$	T-100 routing fraction: share of country c 's US-bound passengers landing at city v (see A.2). By construction $\sum_v f_{c,v}^T = 1$.
ϕ_c	Country-specific 2026 growth factor (see A.3)
g_c	Background (non-WC) trend component of ϕ_c , representing non-WC travel growth from 2024 to 2026
$m_{c,v}$	Number of group-stage matches that country c plays at US venue v
M_c	Total group-stage matches for country c across all WC venues ($M_c = 3$ for all qualified teams)
$\omega_{c,v}$	Schedule-driven routing fraction for WC fans from country c to US venue v (see A.4)
$N_{c,v}$	Projected June 2026 arrivals from country c to venue city v (model-tier specific)

A.2 T-100 routing fraction $f_{c,v}^T$

The T-100 routing fraction gives the proportion of country c 's US-bound passengers that land at each venue city v :

$$f_{c,v}^T = \frac{T_{c,v}}{\sum_{v'=1}^{11} T_{c,v'}}$$

where $T_{c,v}$ is the mean number of June passengers from country c to US city v over 2023–2025 (BTS T-100 Form 41 data). By construction, $\sum_v f_{c,v}^T = 1$ for every country c . These fractions are held fixed across all three model tiers; only the total volume $A_c \cdot \phi_c$ changes between M1, M2, and M3.

A.3 Growth factor ϕ_c and its components

The growth factor

$$\phi_c = \frac{V_{c,2026}}{V_{c,2024}}$$

scales June 2024 baseline arrivals to projected June 2026 levels, where $V_{c,t}$ denotes projected total June visitors from country c in year t . Three country classes are distinguished:

1. **Top-12 inbound markets** (Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South Korea, UK) — NTTTO country-specific projections:

$$\phi_c = \frac{A_{c,2026}^{\text{NTTTO}}}{A_{c,2024}}$$

The World Cup effect is already embedded in the NTTTO forecast. The background (non-WC) trend component g_c is derived from the pre-tournament NTTTO trend excluding the tournament increment.

2. **Other WC-qualified nations** — a uniform World Cup growth factor is applied:

$$\phi_c = \phi^{\text{WC}} = \frac{85,017}{72,390} = 1.174$$

reflecting the aggregate fan-travel surge for smaller qualified markets (NTTTO 2025 report). The background trend component is $g_c = \phi^{\text{bg}} = 1.077$, so the WC-specific fan increment is $\phi_c - g_c = 0.097$.

3. **Non-qualified nations** — no WC adjustment is made:

$$\phi_c = g_c = \phi^{\text{bg}} = 1.077$$

where ϕ^{bg} is the data-driven background growth rate estimated from COR and T-100 data (June 2023–2025). By construction, $\phi_c - g_c = 0$ for non-qualified countries.

A.4 Schedule-driven routing fraction $\omega_{c,v}$

For WC-qualified nation c , the fan stream is routed proportionally to the share of its group-stage matches played at each US venue:

$$\omega_{c,v} = \frac{m_{c,v}}{M_c} = \frac{m_{c,v}}{3} \quad (\text{US venues only})$$

If country c plays no matches at US venue v then $\omega_{c,v} = 0$. Matches played at Canadian or Mexican venues contribute to M_c but not to any US $\omega_{c,v}$, so:

$$\sum_{v=1}^{11} \omega_{c,v} = \frac{(\text{US matches of } c)}{3} \leq 1$$

with strict inequality for nations playing one or more matches outside the US. For non-qualified nations no WC fan stream exists and $\omega_{c,v}$ is not used.

A.5 Travel decomposition for M3

Model 3 separates total projected 2026 arrivals from country c into a background stream and a WC fan stream:

$$N_c^{\text{WC}} = A_c \cdot \max(0, \phi_c - g_c) \quad N_c^{\text{bg}} = A_c \cdot g_c$$

Note that $N_c^{\text{WC}} + N_c^{\text{bg}} = A_c \cdot \phi_c$, so the decomposition is exact and exhaustive. For non-qualified countries $\phi_c = g_c$, giving $N_c^{\text{WC}} = 0$. For typical WC-qualified nations the fan increment represents approximately $(1.174 - 1.077)/1.174 \approx 8.3\%$ of total projected 2026 arrivals.

A.6 Travel flow $N_{c,v}$ by model tier

M1 — Baseline. Raw June 2024 arrivals distributed by T-100 routing fractions; no growth factor applied.

$$N_{c,v}^{M1} = A_c \cdot f_{c,v}^T$$

M2 — WC-adjusted. Projected 2026 arrivals (incorporating the full growth factor ϕ_c) distributed by T-100 routing fractions. All WC fan travel is assigned to US cities regardless of whether matches are played in Canada or Mexico.

$$N_{c,v}^{M2} = A_c \cdot \phi_c \cdot f_{c,v}^T$$

M3 — Schedule-driven. The background stream is T-100 routed; the WC fan stream is routed by the match schedule:

$$N_{c,v}^{M3} = \underbrace{N_c^{\text{bg}} \cdot f_{c,v}^T}_{\text{background stream}} + \underbrace{N_c^{\text{WC}} \cdot \omega_{c,v}}_{\text{WC fan stream}} = A_c \left[g_c \cdot f_{c,v}^T + (\phi_c - g_c) \cdot \omega_{c,v} \right]$$

For non-qualified nations $\phi_c = g_c$, so the fan stream vanishes and $N_{c,v}^{M3} = A_c \cdot g_c \cdot f_{c,v}^T$.

A.7 Why $N_{c,v}^{M3} \leq N_{c,v}^{M2}$ for US cities

M2 routes the entire WC increment $(\phi_c - g_c)$ through T-100 fractions, which sum to 1 over US cities. M3 routes the same increment through schedule fractions, which sum to ≤ 1 over US cities (strictly < 1 whenever country c plays matches in Canada or Mexico). Therefore:

$$\sum_v N_{c,v}^{M3} = A_c \left[g_c \underbrace{\sum_v f_{c,v}^T}_{=1} + (\phi_c - g_c) \underbrace{\sum_v \omega_{c,v}}_{\leq 1} \right] \leq A_c \cdot \phi_c = \sum_v N_{c,v}^{M2}$$

The gap equals the fan travel redirected to non-US WC venues (Canada, Mexico), which is why the M3 US-city total lies below M2 for all qualified nations with at least one non-US match. Importation probability heatmaps for M1 and M2 are shown in Supplementary Figures S1 and S2; these complement the M3 heatmap in Figure 1 and the three-model comparison in Figure 2 of the main text.

Appendix B. Mathematical specification of the diaspora network extension

B.1 Notation

Symbol	Definition
$D_{c,v}$	Number of US residents born in country c living in metro area v (ACS 5-year estimates 2019–2023, Table B05006)
FB_v	Total foreign-born population of metro area v (row total of Table B05006)
$\kappa_{c,v}$	Diaspora concentration index: share of metro v 's foreign-born population born in country c
$\Omega_{c,v,d}^A$	Expected number of infected arrivals from c at city v that enter the co-national network (Mechanism A)
$P^A(\geq 1)_{v,d}$	Probability that at least one infected arrival enters any co-national network in city v
v_{match}	WC venue city where country c plays
v_{home}	Diaspora hub city (may differ from v_{match})
$\omega_{c,v}$	Schedule-driven routing fraction: share of c 's group-stage matches played at US venue v (Appendix A.4); zero if c does not play at v
$\Omega_{c,v_h,d}^B$	Secondary seeding index for hub city v_h , summed over all match venues where c plays (Mechanism B)

B.2 Diaspora concentration index $\kappa_{c,v}$

For each source country c and venue metro area v :

$$\kappa_{c,v} = \frac{D_{c,v}}{FB_v}$$

$\kappa_{c,v} \in [0, 1]$ measures how concentrated the country- c community is *within* the immigrant population of city v — the relevant social network for co-national mixing. Summed over all countries of birth, $\sum_c \kappa_{c,v} = 1$ for each v ; because the model includes only the source countries relevant to the five diseases, the $\kappa_{c,v}$ for the modeled countries sum to less than one in each city.

B.3 Mechanism A: local social mixing

Let $X_{c,v,d} \sim \text{Poisson}(\lambda_{c,v,d})$ be the number of infected arrivals from country c at city v . Under the proportional-mixing assumption, each arrival independently enters the co-national network with probability $\kappa_{c,v}$. By the Poisson thinning theorem [1], the thinned count is:

$$Y_{c,v,d} \sim \text{Poisson}(\lambda_{c,v,d} \cdot \kappa_{c,v})$$

We define the community-weighted importation intensity:

$$\Omega_{c,v,d}^A = \lambda_{c,v,d} \cdot \kappa_{c,v} = \lambda_{c,v,d} \cdot \frac{D_{c,v}}{FB_v}$$

Because $Y_{c,v,d}$ values are independent Poisson random variables, the city-level aggregate $\Omega_{v,d}^A = \sum_c \Omega_{c,v,d}^A$ is also a valid Poisson rate. The probability that at least one infected

arrival enters any co-national network in city v is:

$$P^A(\geq 1)_{v,d} = 1 - e^{-\Omega_{v,d}^A}$$

This is directly comparable to the main-model probability $P(\geq 1)_{v,d} = 1 - e^{-\Lambda_{v,d}}$. Since $\kappa_{c,v} \leq 1$ for all c, v , it follows that $\Omega_{v,d}^A \leq \Lambda_{v,d}$ and therefore $P^A(\geq 1)_{v,d} \leq P(\geq 1)_{v,d}$ always: the diaspora-network probability is a lower bound on the arrival probability, never exceeding it.

B.4 Mechanism B: diaspora return seeding

US-resident diaspora members from country c living in hub city v_{home} may travel domestically to v_{match} (where country c plays a US match), be exposed to infected co-nationals within the local network that Mechanism A already quantifies, and seed transmission upon return. The secondary seeding index reuses the schedule-driven routing fraction $\omega_{c,v}$ (Appendix A.4) to restrict this pathway to venues where c actually plays, and the local exposure hazard $\Omega_{c,v,d}^A$ (Section B.3) rather than the raw arrival rate $\lambda_{c,v,d}$:

$$\Omega_{c, v_{\text{home}}, d}^B = \sum_{v_{\text{match}} \neq v_{\text{home}}} \omega_{c, v_{\text{match}}} \cdot \Omega_{c, v_{\text{match}}, d}^A \cdot \kappa_{c, v_{\text{home}}}$$

Because $\omega_{c,v} \leq 1$ and $\kappa_{c,v} \leq 1$ for all c, v , and $\Omega_{c,v,d}^A$ is already a well-scaled Poisson rate (Section B.3), Ω^B remains on the same scale as Ω^A rather than growing unboundedly, unlike a specification that instead scales a raw diaspora headcount by an assumed match-attendance probability. Summing over source countries gives the hub-city aggregate:

$$\Omega_{v_{\text{home}}, d}^B = \sum_c \Omega_{c, v_{\text{home}}, d}^B$$

and the corresponding probability that at least one returning diaspora member seeds transmission in the hub city, $P^B(\geq 1)_{v_{\text{home}}, d} = 1 - e^{-\Omega_{v_{\text{home}}, d}^B}$, directly comparable to $P^A(\geq 1)_{v,d}$.

Because $\omega_{c, v_{\text{match}}} = 0$ whenever country c does not play a match at v_{match} (Appendix A.4), Ω^B is defined only for the subset of source countries that are both WC-qualified and scheduled to play at a US venue — 13 of the source countries modeled here meet this condition. A country with substantial $\Omega_{c,v,d}^A$ driven by routine background travel to a city where it is not playing contributes zero to Ω^B at that venue, since Mechanism B specifically models exposure acquired while attending a match, not general travel; that country's local risk is still fully captured by Mechanism A. This is a deliberate scope restriction consistent with the mechanism's premise, not an omission.

The match venue v_{match} ranges over the 11 host cities, while the hub city v_{home} ranges over both the 11 host cities and 15 major non-host metropolitan areas with large immigrant populations (drawn from the same ACS B05006 country-of-birth data), so that secondary seeding into non-venue cities is captured. The 15 non-host hub metropolitan areas are Austin, Charlotte, Chicago, Denver, Detroit, Las Vegas, Minneapolis, Orlando, Phoenix, Portland, Sacramento, San Antonio, San Diego, Tampa, and Washington, DC.

Table S1. Sensitivity of expected importations to the mild-symptom travel probability p_d .

Because p_d enters the model multiplicatively ($\lambda_{c,v,d} = N_{c,v} \cdot \rho_d \cdot C_{c,d}/P_c \cdot p_d$), fixing ρ_d at its literature midpoint and varying p_d produces importation totals that scale exactly as p_d/p_{mid} . The table below reports these relative values (Λ_{rel} , normalised to 1.00 at p_{mid}) and the exact fold-change $p_{\text{max}}/p_{\text{min}}$ for all five modelled diseases. Absolute 11-city importation counts at each p_d level are available in the analysis code at https://github.com/jdiestra1977/FIFA_worldCup_2026_risk.

Disease	p_{min}	p_{mid}	p_{max}	$\Lambda(p_{\text{min}})$	$\Lambda(p_{\text{mid}})$	$\Lambda(p_{\text{max}})$	Fold
Dengue	0.30	0.50	0.70	29.70	49.50	69.29	2.3×
Malaria	0.10	0.30	0.50	6.08	18.25	30.42	5.0×
Measles	0.02	0.06	0.10	0.02	0.05	0.08	5.0×
Pertussis	0.50	0.70	0.90	1.49	2.08	2.68	1.8×
Influenza	0.30	0.50	0.70	1.19	1.98	2.78	2.3×

Λ = 11-city total expected importations (M3 schedule-driven model), computed deterministically with ρ_d fixed at its literature midpoint $(\rho_{\text{min}} + \rho_{\text{max}})/2$. Fold = $\Lambda(p_{\text{max}})/\Lambda(p_{\text{min}}) = p_{\text{max}}/p_{\text{min}}$ (exact). Malaria and measles carry the largest sensitivity to p_d ; pertussis the smallest.

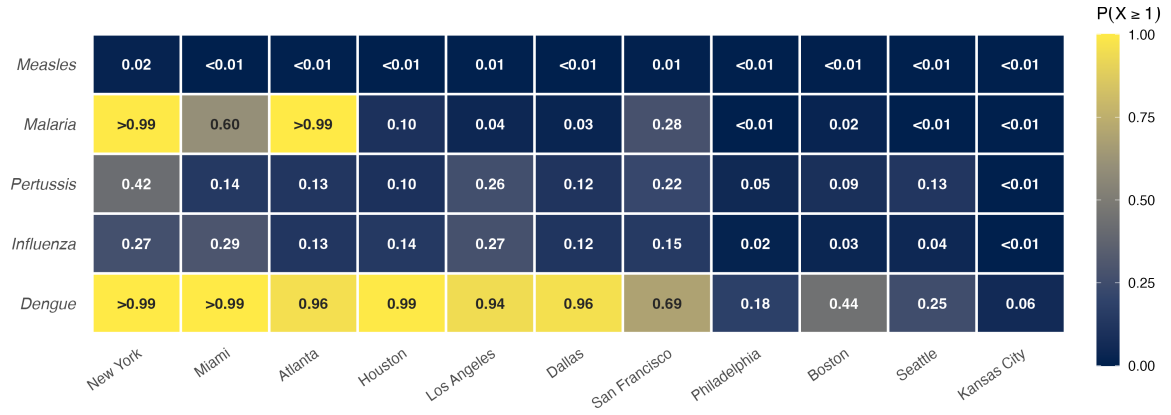


Figure S1. Importation probability under the Baseline model (M1), June 2024 (no World Cup adjustment). Same layout as Figure 1 in the main text. M1 represents the pre-tournament counterfactual: typical June travel volumes without the World Cup surge. Dengue $P(\geq 1) > 0.99$ is already present at Miami, New York, and Houston under baseline conditions, confirming that the World Cup amplifies but does not create the dengue importation threat.

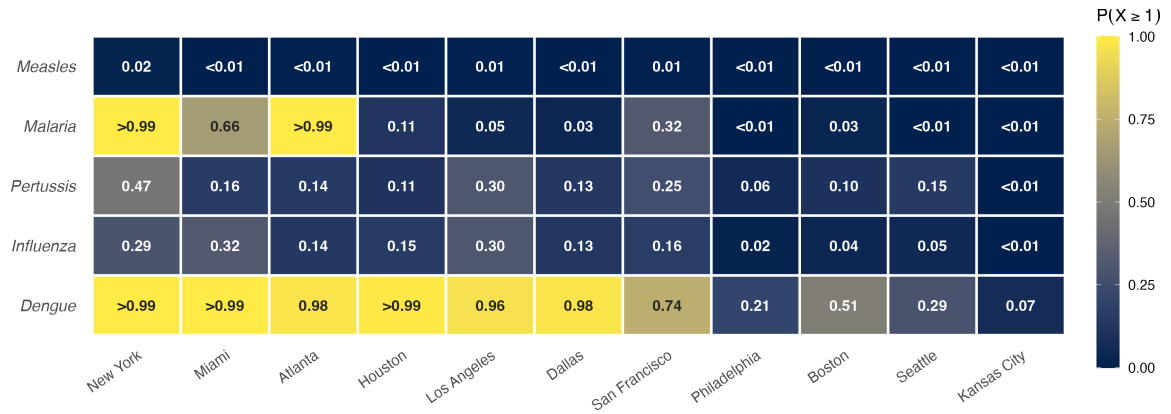


Figure S2. Importation probability under the WC-adjusted model (M2), June 2026. Same layout as Figure 1 in the main text. M2 applies World Cup growth factors to national arrivals but distributes all travel to US cities via T-100 routing fractions, including the fan increment for teams playing in Canada and Mexico. M2 values therefore represent a slight overestimate relative to M3 for US cities.

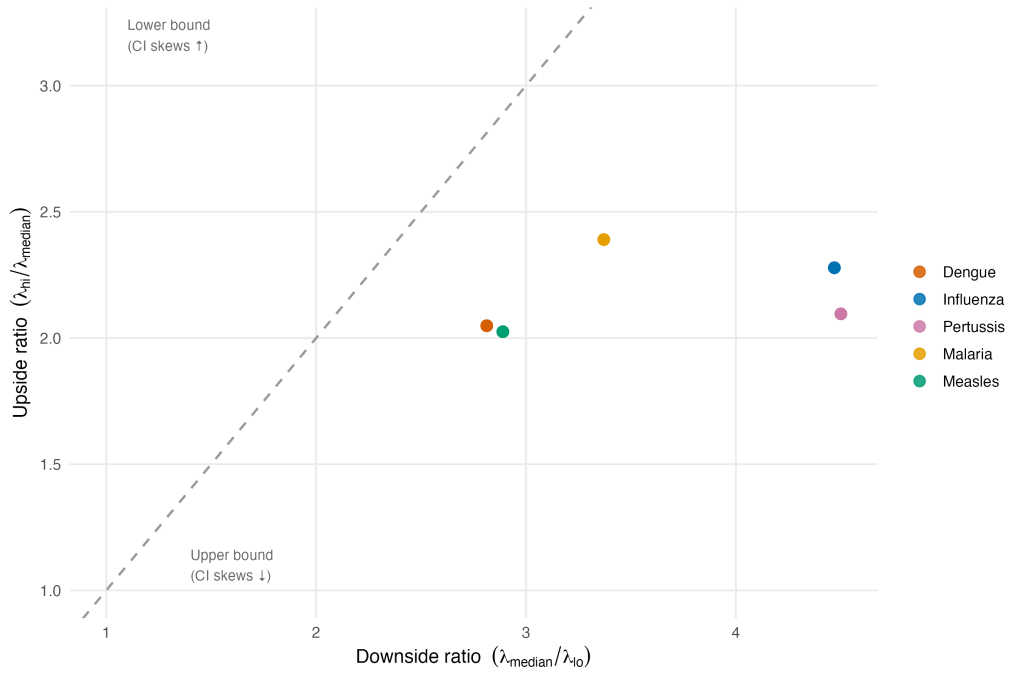


Figure S3. Direction of Monte Carlo uncertainty by disease. Each point represents one disease–city combination. X-axis: downside ratio ($\lambda_{\text{median}}/\lambda_{\text{lo}}$), measuring how far the lower CI bound lies below the median. Y-axis: upside ratio ($\lambda_{\text{hi}}/\lambda_{\text{median}}$), measuring how far the upper CI bound lies above the median. Points above the diagonal have CIs that extend further upward; points below the diagonal have CIs that extend further downward, placing the central estimate closer to the upper plausible bound. All five diseases fall below the diagonal: uncertainty is asymmetrically larger on the downside for every disease. Dengue and measles sit closest to the diagonal (downside ratio ≈ 2.8), while malaria, influenza, and pertussis show larger downside ratios (4.3–4.8), indicating their medians are more conservative upper estimates.



Figure S4. Geographic distribution of the 16 FIFA World Cup 2026 stadiums across the United States, Canada, and Mexico. Colours indicate host nation. This analysis focuses on the 11 US venues (blue); Canadian (Toronto, Vancouver) and Mexican (Mexico City, Monterrey, Guadalajara) venues are shown for context. The East Coast cluster (New York, Philadelphia, Boston) is the densest; Kansas City and Seattle are the most geographically isolated US venues.

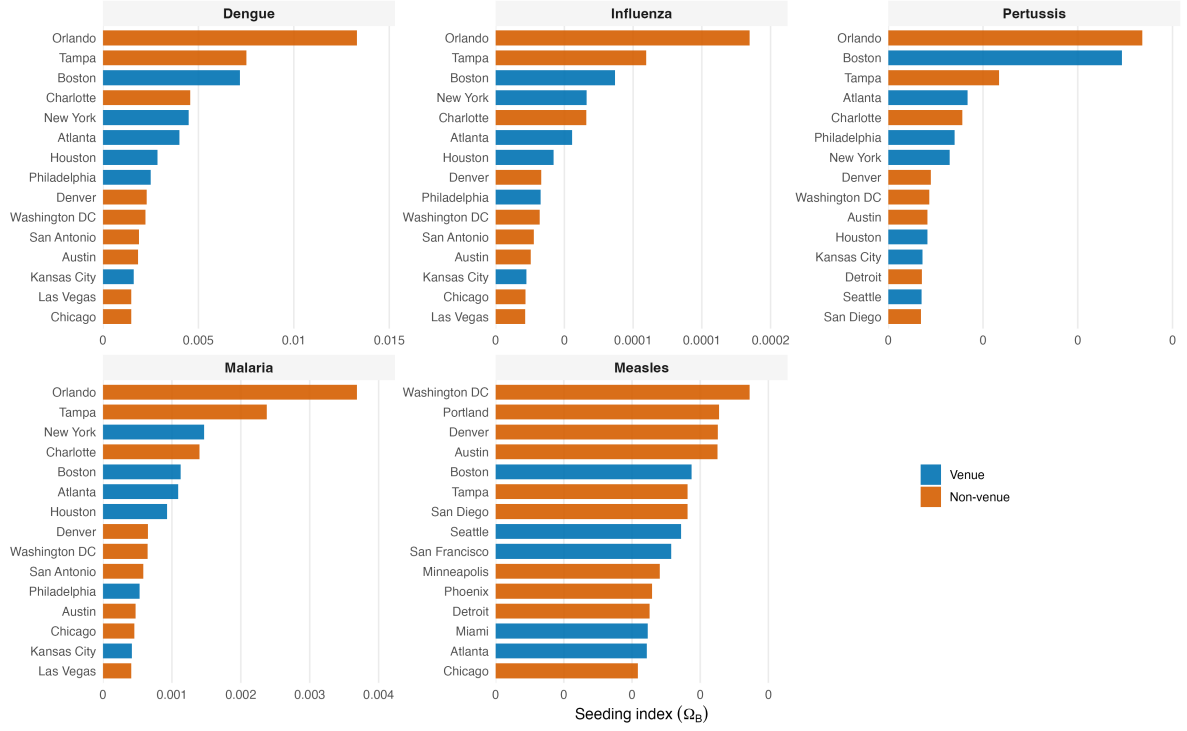


Figure S5. Secondary seeding risk index (Ω^B) for diaspora hub cities, by disease (Mechanism B). Bars show the top 15 hub metropolitan areas per disease, ranked by Ω^B summed across all source countries and match venues, and colored by whether the hub is a World Cup host city or a non-host city. Hubs are the home cities to which diaspora members return after attending matches. Ω^B is defined only for source countries that are both WC-qualified and scheduled to play at a US venue (13 of the modeled source countries), since the mechanism specifically requires a match to attend; a country with high raw importation intensity but no US match, or whose risk arrives mainly via routine background travel rather than the match-attendance pathway, can therefore rank low or absent here despite dominating the raw source-country rankings (main text Figure 4).

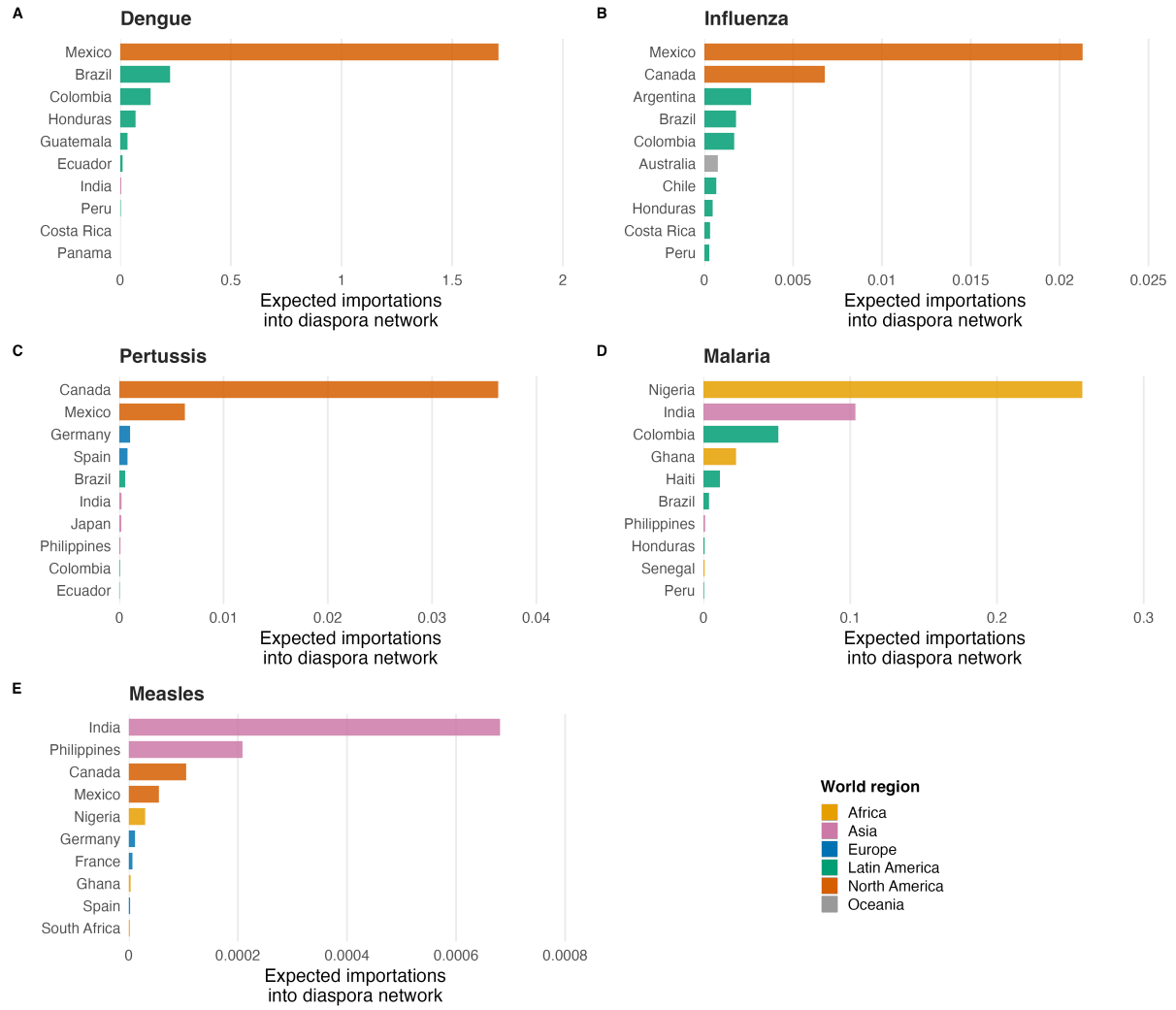


Figure S6. Top 10 source-country contributors to diaspora-network importation intensity (Ω^A) per disease. Bars show Ω^A for each source country, summed across the 11 host cities; colors denote world region (Canada and Mexico grouped as North America). The corresponding raw- λ ranking is shown in Figure 4 of the main text.

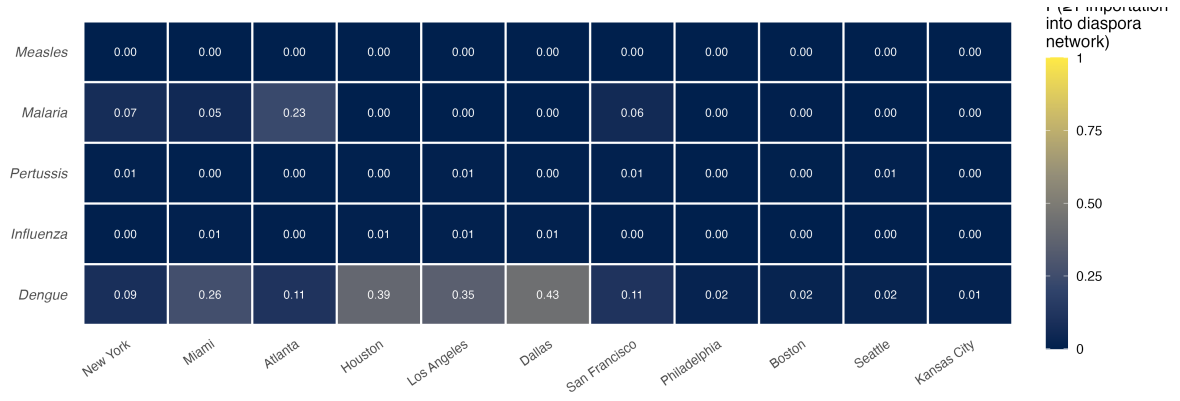


Figure S7. Diaspora network importation probability ($P^A(\geq 1)$) for all five diseases at 11 US host cities. Cells show $P^A(\geq 1)_{v,d}$, the probability that at least one imported case enters a co-national diaspora network at each host city. The dengue and malaria rows correspond to Panel A of Figure 5 in the main text.

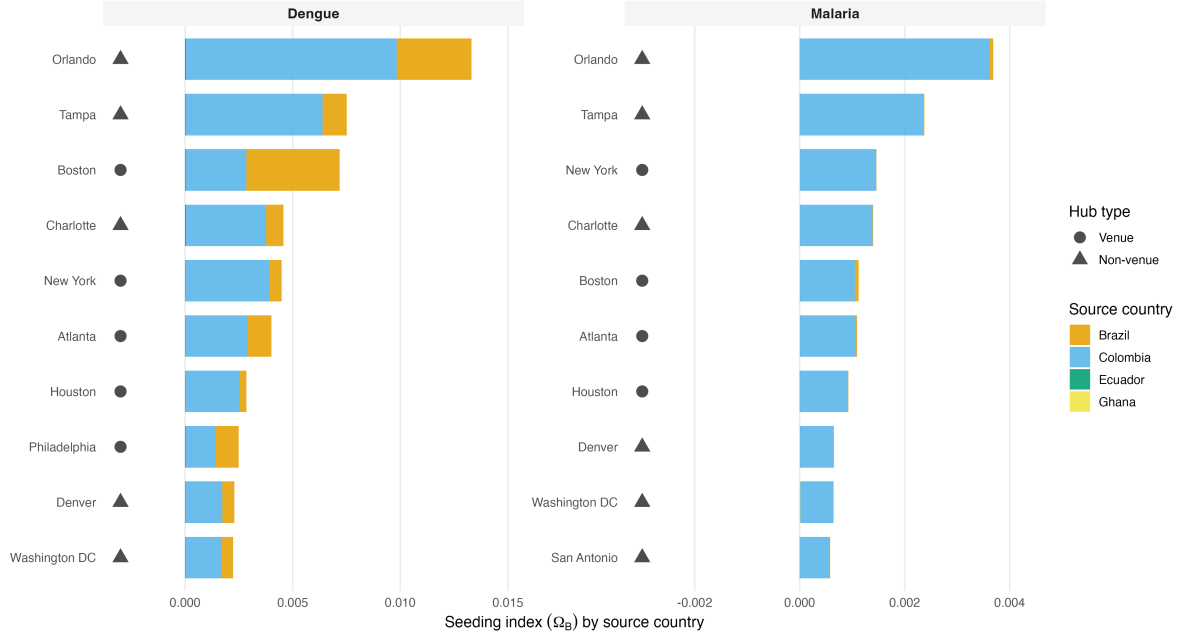


Figure S8. Source-country contributions to Mechanism B secondary seeding index (Ω^B) for dengue and malaria. Bars show the source-country contributions to Ω^B at each of the top hub cities per disease, ranked by total Ω^B . Dot shape distinguishes World Cup venue cities (circle) from non-venue metropolitan areas (triangle). Because Mechanism B specifically requires a diaspora member to travel to a match involving their home country, it is restricted to the subset of source countries that are both WC-qualified and scheduled to play at a US venue; for both dengue and malaria, Colombia and Brazil are the only such countries with incidence high enough to register, so they drive secondary seeding at every top hub. Countries with much larger raw arrival intensities for these diseases (e.g., Nigeria for malaria) do not appear here because their risk arrives via routine background travel to cities where their team is not playing, a pathway Mechanism A already captures but that falls outside Mechanism B’s match-attendance pathway (see main text Limitations and Appendix B.4). Non-venue hubs (Orlando, Tampa, Charlotte, Denver, Washington, DC) rank alongside host cities because of their large Colombian and Brazilian diaspora concentrations.

References

- [1] Ross SM. Introduction to Probability Models. 10th ed. Burlington, MA: Academic Press; 2010.